BAS-ACC-05 Access Node BGP Route Churn, OSPF Adjacency Loss, ISIS LSP Flooding Resolution SOP for Huawei CLI Dialect

Node: BAS-ACC-05 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_6000

Description

BAS-ACC-05 Access Node is experiencing a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure domain. This is due to the Huawei Access Node's BGP routing protocol's sensitivity to OSPF and ISIS LSP flooding, which can be triggered by various factors such as link failures, routing table corruption, or high CPU utilization. The OSPF adjacency loss can further exacerbate the issue by causing the node to re-converge and re-establish adjacencies, leading to a BGP route churn. The ISIS LSP flooding can also contribute to the issue by overwhelming the node's CPU and memory resources, leading to instability and potential crashes. To resolve this issue, it is essential to identify and rectify the root cause of the problem, which may involve investigating link failures, routing table corruption, high CPU utilization, or other factors that may be contributing to the issue. This SOP outlines the procedures for diagnosing and resolving the issue on the Huawei Access Node.

Pre-requisite Checks

- Verify that the BAS-ACC-05 Access Node is reachable and accessible via the network.

- Check the redundant link status to ensure that it is operational and not contributing to the issue.

- Dispatch remote hands to the Basavanagudi POP to perform physical checks and ensure that the node's power supply and cooling systems are functioning correctly.

Potential Root Causes

Cause	Probability
BGP routing protocol misconfiguration	High
OSPF adjacency loss due to link failure	Medium
ISIS LSP flooding due to high CPU utilization	Low

Diagnostic Commands & Expected Outputs

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Command: display bgp routing-table, Expected Output: show the current BGP routing table, including the AS path and next hop for each prefix, Description: This command will help identify any BGP route churn and provide information on the AS path and next hop for each prefix.

Command: display ospf neighbor, Expected Output: show the current OSPF neighbor status, including the neighbor ID, local RID, and state, Description: This command will help identify any OSPF adjacency loss
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and provide information on the neighbor status, including the neighbor ID, local RID, and state.

Command: display isis lsp, Expected Output: show the current ISIS LSP list, including the LSP ID, sequence number, and checksum, Description: This command will help identify any ISIS LSP flooding and provide information on the LSP list, including the LSP ID, sequence number, and checksum.
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Resolution Steps

Step 1: Step: Verify BGP routing protocol configuration, Command: display bgp configuration, Description: Verify that the BGP routing protocol is properly configured, including the AS number, router ID, and neighbor IP address.

Step 2: Step: Reset OSPF adjacency, Command: reset ospf neighbor, Description: Reset the OSPF adjacency to ensure that the node reconnects to its neighbors and re-establishes adjacencies.

Step 3: Step: Clear ISIS LSP, Command: clear isis lsp, Description: Clear the ISIS LSP to prevent LSP flooding and ensure that the node's CPU and memory resources are not overwhelmed.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: show the updated BGP routing table, including the AS path and next hop for each prefix, Description: Verify that the BGP route churn has been resolved and that the AS path and next hop for each prefix are correctly updated.

Command: display ospf neighbor, Expected Output: show the updated OSPF neighbor status, including the neighbor ID, local RID, and state, Description: Verify that the OSPF adjacency has been re-established and that the neighbor status, including the neighbor ID, local RID, and state, is correctly updated.

Command: display isis lsp, Expected Output: show the updated ISIS LSP list, including the LSP ID, sequence number, and checksum, Description: Verify that the ISIS LSP flooding has been cleared and that the LSP list, including the LSP ID, sequence number, and checksum, is correctly updated.
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Roll-back Procedures

- Step: Restore original BGP routing protocol configuration, Command: undo bgp configuration, Description: Restore the original BGP routing protocol configuration to ensure that the node's BGP routing protocol is properly configured.

- Step: Reset OSPF adjacency, Command: reset ospf neighbor, Description: Reset the OSPF adjacency to ensure that the node reconnects to its neighbors and re-establishes adjacencies.

- Step: Clear ISIS LSP, Command: clear isis lsp, Description: Clear the ISIS LSP to prevent LSP flooding and ensure that the node's CPU and memory resources are not overwhelmed.